

MELBOURNE MOORABBIN, AustraliaWMO: **948700**Lat: **37.9800S**Lon: **145.0964E**Elev: **42**StdP: **14.67**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.9	39.2	33.3	27.9	50.7	35.0	29.8	49.5	29.5	56.2	27.3	56.4	2.6	0	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	94.4	66.7	89.4	65.5	84.8	64.9	70.7	82.9	69.0	81.0	67.5	79.2	15.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	101.8	73.3	65.5	94.2	72.7	63.6	88.1	71.5	34.6	82.7	33.2	80.7	31.9	79.2	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	23.2	20.8	DB	32.4	103.7	2.0	4.7	31.0	107.0	29.8	109.8	28.7	112.4	27.3	115.8	(4)
(5)			WB	31.8	73.9	2.0	2.3	30.4	75.6	29.2	76.9	28.1	78.2	26.7	79.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	67.9	68.1	64.9	59.3	55.0	50.8	50.1	51.4	54.1	57.5	61.2	64.5	(6)	
	DBStd	8.27	6.91	6.32	6.09	5.11	4.33	3.70	3.57	3.91	4.84	5.65	6.26	6.57	(7)	
	HDD50	118	0	0	0	1	5	33	42	27	9	1	0	0	(8)	
	HDD65	2773	41	32	78	186	312	425	461	421	329	245	153	90	(9)	
	CDD50	3287	555	507	461	279	160	58	46	71	133	234	335	448	(10)	
	CDD65	466	130	119	74	14	1	0	0	0	3	13	38	74	(11)	
	CDH74	5011	1374	1123	724	136	3	0	0	0	17	226	495	913	(12)	
(13)	CDH80	2185	687	498	285	27	0	0	0	0	1	63	187	437	(13)	
(14)	Wind	WSAvg	10.2	10.8	10.3	9.9	8.7	9.4	9.2	10.6	10.7	10.8	10.6	10.8	(14)	
(15) Precipitation	PrecAvg	28.5	1.9	1.6	1.9	2.5	3.0	2.2	2.6	2.8	2.6	2.8	2.4	2.3	(15)	
	PrecMax	38.0	5.4	8.3	5.1	6.2	6.3	5.5	4.1	5.0	4.8	6.5	7.1	4.9	(16)	
	PrecMin	17.5	0.0	0.0	0.5	0.3	0.9	0.6	0.7	0.8	1.0	0.9	0.2	0.1	(17)	
	PrecStd	5.0	1.3	1.9	1.1	1.3	1.3	1.1	0.9	0.9	1.0	1.3	1.5	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.2	97.9	95.2	83.9	72.6	64.7	63.7	69.4	76.2	87.1	91.4	98.5	(19)	
		MCWB	69.4	67.5	67.0	62.1	58.0	55.4	53.3	56.4	58.4	61.8	64.7	66.3	(20)	
	2%	DB	94.3	91.7	87.2	77.7	68.5	61.8	60.6	64.8	71.3	80.1	85.6	90.9	(21)	
		MCWB	67.1	66.9	65.3	60.7	57.3	54.3	52.3	53.6	57.2	60.6	63.9	64.9	(22)	
	5%	DB	87.7	86.6	81.9	73.4	65.7	59.3	58.2	61.6	67.4	74.8	79.6	84.1	(23)	
		MCWB	66.9	66.3	64.1	59.8	56.1	53.0	50.9	52.3	55.5	58.9	62.3	63.9	(24)	
	10%	DB	81.0	80.8	76.6	69.4	62.9	57.4	56.4	58.7	63.8	69.6	73.9	77.6	(25)	
		MCWB	66.3	66.0	63.7	59.0	55.3	52.0	50.1	50.9	54.1	57.1	61.2	63.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	72.6	70.2	64.9	61.1	57.6	55.6	57.5	61.3	64.8	68.9	70.3	(27)	
		MCDB	86.0	82.8	82.8	76.4	67.0	61.1	61.0	65.9	70.5	77.6	82.3	82.9	(28)	
	2%	WB	71.1	70.3	68.0	63.0	59.0	55.5	53.4	54.9	58.7	62.3	66.2	67.9	(29)	
		MCDB	84.5	81.0	79.3	71.3	65.1	59.7	58.2	62.8	69.0	75.0	78.9	81.4	(30)	
	5%	WB	69.1	68.8	66.2	61.5	57.4	54.0	52.0	53.0	56.5	60.2	64.3	66.0	(31)	
		MCDB	81.7	79.8	77.3	69.9	63.7	58.2	56.6	59.5	65.4	71.7	76.0	78.9	(32)	
	10%	WB	67.2	67.2	64.5	59.9	55.9	52.7	50.9	51.7	54.6	58.0	62.2	64.2	(33)	
		MCDB	79.0	77.9	74.6	68.0	61.7	56.8	55.3	57.4	62.1	67.8	72.2	76.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	18.1	17.2	15.9	12.8	11.9	11.3	12.4	14.4	16.8	17.3	18.3	(35)	
		MCDBR	33.5	30.7	28.1	24.3	17.3	14.9	14.0	17.3	20.8	28.7	30.2	33.7	(36)	
		MCWBR	10.6	10.1	10.2	10.7	9.2	9.3	8.4	9.1	10.3	12.5	12.0	11.5	(37)	
	5% WB	MCDBR	24.8	22.2	21.2	19.5	14.9	12.4	11.9	15.4	18.4	24.2	24.7	25.0	(38)	
		MCWBR	10.3	9.9	9.6	9.9	9.0	8.5	7.8	8.9	10.2	12.0	11.2	11.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.355	0.359	0.335	0.332	0.316	0.306	0.307	0.318	0.341	0.341	0.350	0.344	(40)	
	taud		2.503	2.502	2.568	2.548	2.577	2.595	2.580	2.540	2.472	2.489	2.493	2.525	(41)	
	Ebn at Noon		310	301	296	277	262	256	263	277	289	303	310	315	(42)	
	Edh at Noon		36	34	30	26	22	20	22	26	32	34	36	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2257	1971	1566	1087	716	559	625	885	1265	1697	1997	2233	(44)	
(45)	RadStd		139	102	88	62	41	45	33	56	88	132	115	126	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.31	N/A	N/A	+2.11	+0.68	N/A	-32	-256	+405	+169
Regional (37 neighbors)	+0.92	N/A	N/A	+1.51	N/A	N/A	-32	-234	+313	+104

Nomenclature: See separate page